

From Mean Field Games to Navier–Stokes Equations

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- ① Navier–Stokes Equations
- ② Mean Field Games (MFG)
- ③ Connecting Navier–Stokes to MFG
- ④ Fixed Point Formulation
- ⑤ Existence via Schauder's Theorem
- ⑥ Conclusion

Navier–Stokes Equations

The Navier–Stokes equations model the motion of viscous fluids by describing how fluid velocity is affected by pressure, viscosity, and external forces.

The incompressible Navier–Stokes equations are

$$\begin{cases} (\partial_t + v \cdot \nabla)v = \frac{\nu \Delta v - \nabla p + f}{\rho}, \\ \nabla \cdot v = 0, \\ v(0, x) = v_0(x) \end{cases}$$

- $v(t, x)$: velocity field
- $\nu > 0$: dynamic viscosity
- $p(t, x)$: pressure
- $f(t, x)$: external force per unit volume
- $\rho > 0$: density

Compressible Navier–Stokes Equations

For compressible fluids, the density ρ becomes a function of space and time. The compressible Navier–Stokes equations for isentropic fluids are

$$\left\{ \begin{array}{l} (\partial_t + v \cdot \nabla)v = \frac{\nu \Delta v - \nabla p + f}{\rho}, \\ \partial_t \rho + \nabla \cdot (\rho v) = 0, \\ v(0, x) = v_0(x), \\ \rho(0, x) = \rho_0(x). \end{array} \right.$$

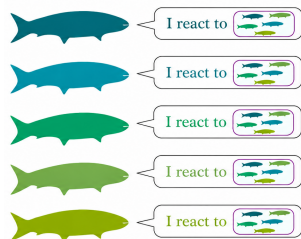
The fluid density $\rho(t, x)$ evolves together with the velocity field $v(t, x)$ over time. The pressure $p[\rho](t, x)$ is affected by the fluid density.

Mean Field Games

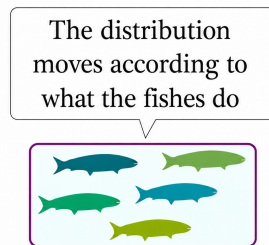
Mean Field Games model stochastic differential games in the limit of large populations, where each individual player responds to the aggregate distribution of all players.

The theory is built around two coupled equations:

- a Hamilton–Jacobi–Bellman equation for each player's optimization problem;
- a Fokker–Planck–Kolmogorov equation for the evolution of the population distribution.



Hamilton-Jacobi-Bellman



Fokker-Planck-Kolmogorov

Stochastic HJB Equation

Stochastic Time Evolution: $dX_s = a(s)ds + dB_s; \quad X_t = x$

Cost Functional: $J(x, t, a) = \int_t^T \left(\frac{1}{2} |a(s)|^2 - p(X_s, s) \right) ds + h(X_T)$

Value Function: $u(x, t) = \inf_{a(\cdot)} \mathbb{E}[J(x, t, a)]$

Hamiltonian: $H(x, \nabla u) = \inf_a \left[\frac{1}{2} |a|^2 + \nabla u \cdot a - p \right] = -\frac{1}{2} |\nabla u|^2 - p$

Stochastic HJB Equation $-\partial_t u = H(x, \nabla u) + \frac{1}{2} \Delta u = -\frac{1}{2} |\nabla u|^2 - p + \frac{1}{2} \Delta u$

$$u(x, T) = h(x)$$

The Mean Field Game System

The MFG system considered couples a backward HJB equation with a forward Fokker–Planck (convection–diffusion) equation:

$$\begin{cases} -\partial_t u - \frac{1}{2} \Delta u + \frac{1}{2} |\nabla u|^2 = -p \\ \partial_t \rho - \frac{1}{2} \Delta \rho - \nabla \cdot (\rho \nabla u) = 0 \\ u(T, x) = h(x) \\ \rho(0, x) = \mu(x) \end{cases}$$

Where:

- $u(t, x)$: value function for a representative player
- $\frac{1}{2} |\nabla u|^2$ is the Hamiltonian of this system
- $\rho(t, x)$: population density
- $-\nabla u$: optimal feedback control
- $p(x, t)$ represents the effect of density on the objective functional

From Navier–Stokes to Mean Field Games

The paper modifies the compressible Navier–Stokes system in the following ways:

1. **Zero external force and constant viscosity:**

$$f = 0, \quad \nu = \frac{1}{2}$$

2. **Assume** $v_0(x) = \nabla h(x)$

3. **Density–dependent pressure and terminal condition:**

$$\frac{p}{\rho} = p[\rho] \quad \text{and} \quad h = h[\rho]$$

4. **Add reverse diffusive term to continuity equation:**

$$\partial_t \rho + \nabla \cdot (\rho v) = -\frac{1}{2} \Delta \rho$$

5. $\mu(x)$ is the smooth terminal density distribution:

$$\rho(T, x) = \mu(x) \quad \text{and} \quad \mu \in C^\infty$$

Modifying the Navier–Stokes

Modified Navier–Stokes

$$\left\{ \begin{array}{lcl} (\partial_t + v \cdot \nabla)v & = & -\nabla p[\rho] + \frac{1}{2}\Delta v \\ \partial_t \rho + \nabla \cdot (\rho v) + \frac{1}{2}\Delta \rho & = & 0 \\ v(0, x) & = & \nabla h[\rho](x) \\ \rho(T, x) & = & \mu(x) \end{array} \right.$$

Modified Navier–Stokes with Time Reversal $t \mapsto T - t$ and $v = \nabla u$

$$\left\{ \begin{array}{lcl} -\partial_t u + \frac{1}{2}|\nabla u|^2 & = & -p[\rho] + \frac{1}{2}\Delta u \\ -\partial_t \rho + \nabla \cdot (\rho \nabla u) + \frac{1}{2}\Delta \rho & = & 0 \\ u(T, x) & = & h[\rho](x) \\ \rho(0, x) & = & \mu(x) \end{array} \right.$$

Modified Navier–Stokes to Mean Field Games

Modified Navier–Stokes with Time Reversal $t \mapsto T - t$ and $v = \nabla u$

$$\begin{cases} -\partial_t u - \frac{1}{2}\Delta u + \frac{1}{2}|\nabla u|^2 = -p[\rho] \\ \partial_t \rho - \frac{1}{2}\Delta \rho - \nabla \cdot (\rho \nabla u) = 0 \\ u(T, x) = h[\rho](x) \\ \rho(0, x) = \mu(x) \end{cases}$$

Mean Field Games

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Set-Up

The state space is the three-dimensional torus

$$\mathbb{T}^3 = \mathbb{R}^3 / \mathbb{Z}^3.$$

This makes the spatial domain a periodic version of the cube $[0, 1)^3$.



Figure: A wrap-around screen is a one-dimensional analogy for periodic boundary conditions

Norms and Hölder Seminorms

For a function

$$f : [0, T] \times \mathbb{T}^3 \rightarrow S,$$

where S is a Banach space, define

$$|f|_0 = \sup_{(t,x) \in [0,T] \times \mathbb{T}^3} |f(t, x)|.$$

For $\delta_1, \delta_2 \in [0, 1]$, define the Hölder seminorm

$$[f]_{\delta_1, \delta_2} = \sup_{\substack{0 \leq t \neq t' \leq T \\ x \neq x' \in \mathbb{T}^3}} \frac{|f(t, x) - f(t', x')|}{|t - t'|^{\delta_1} + |x - x'|^{\delta_2}}.$$

This measures how smoothly f changes across time and space.

We write

$$f \in C^{m+\delta_1, n+\delta_2}([0, T] \times \mathbb{T}^3, S)$$

if f has m time derivatives and n spatial derivatives, with highest-order derivatives satisfying Hölder regularity.

$$|f|_{m+\delta_1, n+\delta_2} = |f|_{m, n} + [\partial_t^m f]_{\delta_1, \delta_2} + \sum_{|\alpha|=n} [D_x^\alpha f]_{\delta_1, \delta_2} < \infty,$$

where

$$|f|_{m, n} = \sum_{i \leq m} |\partial_t^i f|_0 + \sum_{|\beta| \leq n} |D_x^\beta f|_0.$$

Closed Balls and Compactness

We denote the closed ball of radius r centered at zero by

$$B^{m+\delta_1, n+\delta_2}(r) = \left\{ f \in C^{m+\delta_1, n+\delta_2} : |f|_{m+\delta_1, n+\delta_2} \leq r \right\}.$$

By compact embedding, for $\delta_1 > 0$ and $\delta_2 > 0$:

$$B^{m+\delta_1, n+\delta_2}(r) \text{ is compact in } C^{m, n}([0, T] \times \mathbb{T}^3, S).$$

Why this matters: compactness is the key ingredient needed for Schauder's fixed point theorem.

Probability Distributions on $\mathbb{T}^3$

Let $\mathcal{P} = \mathcal{P}(\mathbb{T}^3)$ be the space of probability distributions on $\mathbb{T}^3$. Since $\mathbb{T}^3$ is compact, probability distributions on $\mathbb{T}^3$ have finite moments.

The paper uses the 1-Wasserstein metric:

$$d_1(\mu, \nu) = \inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathbb{T}^3 \times \mathbb{T}^3} |x - y| d\pi(x, y).$$

Here $\Pi(\mu, \nu)$ is the set of joint distributions on (x, y) whose marginal distributions are μ and ν .

Probability Measure Flows

A probability measure flow is a continuous path

$$\rho \in C^0([0, T], \mathcal{P})$$

with respect to the Wasserstein metric d_1 .

The metric on flows is

$$d_{1T}(\rho_1, \rho_2) = \sup_{0 \leq t \leq T} d_1(\rho_1(t, \cdot), \rho_2(t, \cdot)).$$

For $\delta \in (0, 1)$, we write

$$\rho \in C^\delta([0, T], \mathcal{P})$$

if

$$|\rho|_{1T, \delta} := |\rho|_{1T} + [\rho]_{1T, \delta} < \infty,$$

where

$$|\rho|_{1T} = d_{1T}(\rho, \delta_0) \quad \text{and} \quad [\rho]_{1T, \delta} = \sup_{0 \leq t < t' \leq T} \frac{d_1(\rho(t), \rho(t'))}{|t - t'|^\delta}.$$

Definition of Λ and Λ_0

Let Λ be the collection of functions

$$\lambda : [0, \infty) \rightarrow \mathbb{R}^+$$

satisfying

$$\lambda'(x) > 0 \quad \forall x > 0, \quad \lim_{x \rightarrow \infty} \lambda(x) = \infty.$$

Define

$$\Lambda_0 = \{\lambda \in \Lambda : \lambda(0) = 0\}.$$

Why It Matters: We will bound the changes in $p[\rho]$ and $h[\rho]$ by a function $\lambda_0 \in \Lambda_0$, which as its input the change in density ρ . As the change in density goes to zero, the change in $p[\rho]$ and $h[\rho]$ will go to zero as well.

Main Result: Solvability

We consider the solvability of (ρ, v) for

$$\begin{cases} (\partial_t + v \cdot \nabla)v &= -\nabla p[\rho] + \frac{1}{2}\Delta v, & (t, x) \in [0, T] \times \mathbb{T}^3, \\ \partial_t \rho + \nabla \cdot (\rho v) + \frac{1}{2}\Delta \rho &= 0, & (t, x) \in [0, T] \times \mathbb{T}^3, \\ v(0, x) &= \nabla h[\rho](x), & x \in \mathbb{T}^3, \\ \rho(T, x) &= \mu(x), & x \in \mathbb{T}^3. \end{cases}$$

Theorem (Luo and Song, 2023)

Suppose there exist $\kappa > 0$ and $\lambda_0 \in \Lambda_0$ such that, for all $\rho, \rho_1, \rho_2 \in C^{1/2}([0, T], \mathcal{P})$,

$$|p[\rho]|_{0,2} + |h[\rho]|_4 \leq \kappa, \quad |p[\rho]|_{1/2,2} < \infty,$$

and

$$|p[\rho_1] - p[\rho_2]|_{0,1} + |h[\rho_1] - h[\rho_2]|_4 \leq \lambda_0(d_{1T}(\rho_1, \rho_2)).$$

Then there exists a classical solution (ρ, v) of the system.

The proof has four main steps:

- 1 **Convert to MFG system using $t \mapsto T - t$ and $v = \nabla u$.**
- 2 **Use the MFG system, define a continuous function mapping a compact convex set into itself**
- 3 **Show that a classical solution MFG system is a fixed point of this function**
- 4 **Apply Schauder's Fixed Point Theorem.**

Start of the Proof

Modified Navier–Stokes

$$\begin{cases} (\partial_t + v \cdot \nabla)v = -\nabla p[\rho] + \frac{1}{2}\Delta v \\ \partial_t \rho + \nabla \cdot (\rho v) + \frac{1}{2}\Delta \rho = 0 \\ v(0, x) = \nabla h[\rho](x) \\ \rho(T, x) = \mu(x) \end{cases}$$

After the time reversal $t \mapsto T - t$ and the definition $v = \nabla u$:

Mean Field Game System

$$\begin{cases} -\partial_t u - \frac{1}{2}\Delta u + \frac{1}{2}|\nabla u|^2 = -p[\rho] \\ \partial_t \rho - \frac{1}{2}\Delta \rho - \nabla \cdot (\rho \nabla u) = 0 \\ u(T, x) = h[\rho](x) \\ \rho(0, x) = \mu(x) \end{cases}$$

Defining a Function using the MFG System

The Mean Field Games system is used to define the function: $\Phi^\mu = \Phi_3^\mu \circ \Phi_2 \circ \Phi_1$.

Φ_1 maps density flows $\rho(x, t)$ to the parameters $p[\rho]$ and $h[\rho]$ in the HJB equation.

Φ_2 maps the parameters $p[\rho]$ and $h[\rho]$ to the gradient of the solution of the corresponding stochastic HJB equation $v(x, t) = \nabla u(x, t)$.

Φ_3 maps the velocity field $v(x, t)$ and initial density $\mu(x)$ to the density flow $\rho(x, t)$ using the Fokker–Planck equation.

Operator 1: Φ_1

Φ_1 maps density flows $\rho(x, t)$ to the parameters $p[\rho]$ and $h[\rho]$ in the HJB equation:

$$\Phi_1 : \mathcal{D}_1 \rightarrow \mathcal{D}_2.$$

where $\mathcal{D}_1 = C^{1/2}([0, T], \mathcal{P})$ and $\mathcal{D}_2 = C^{1/2,2}([0, T] \times \mathbb{T}^3, \mathbb{R}) \times C^4(\mathbb{T}^3, \mathbb{R})$

Given a density flow ρ , define

$$\Phi_1[\rho] = (p[\rho], h[\rho])$$

Φ_1 is continuous by the condition that

$$|p[\rho_1] - p[\rho_2]|_{0,1} + |h[\rho_1] - h[\rho_2]|_4 \leq \lambda_0(d_1 \tau(\rho_1, \rho_2)).$$

Operator 2: Φ_2

Φ_2 maps the parameters $p[\rho]$ and $h[\rho]$ to the gradient of the solution of the corresponding stochastic HJB equation $v(x, t) = \nabla u(x, t)$:

$$\Phi_2 : \mathcal{D}_2 \rightarrow \mathcal{D}_3.$$

where $\mathcal{D}_2 = C^{1/2,2}([0, T] \times \mathbb{T}^3, \mathbb{R}) \times C^4(\mathbb{T}^3, \mathbb{R})$ and $\mathcal{D}_3 = C^{1,2}([0, T] \times \mathbb{T}^3, \mathbb{R}^3)$

Given (p, h) , solve the HJB equation

$$\begin{cases} \partial_t u - \frac{1}{2} |\nabla u|^2 + \frac{1}{2} \Delta u - p = 0, & (t, x) \in (0, T) \times \mathbb{T}^3, \\ u(T, x) = h(x), & x \in \mathbb{T}^3. \end{cases}$$

Then define $\Phi_2[p, h] = v = \nabla u$.

$\Phi_2[p, h]$ is well defined by the verification theorem

Continuity of Φ_2

Lemma 1

For any $(p_i, h_i) \in D_2$, let

$$v_i = \Phi_2[p_i, h_i], \quad i = 1, 2,$$

and define

$$M = \max \{ |p_1|_{0,2}, |h_1|_4, |p_2|_{0,2}, |h_2|_4 \}.$$

Then

$$|v_1 - v_2|_0 \leq \lambda(M) (|h_1 - h_2|_4 + |p_1 - p_2|_{0,1}),$$

and

$$|v_1|_{0,1} \leq \lambda(M),$$

for some $\lambda \in \Lambda$.

By the theorem requirements, $|p[\rho]|_{0,2}$ and $|h[\rho]|_4$ are bounded

Therefore, Φ_2 is continuous with respect to p and h

Operator 3: Φ_3^μ

Φ_3 maps the velocity field $v(x, t)$ and initial density $\mu(x)$ to the density flow $\rho(x, t)$:

$$\Phi_3^\mu : \mathcal{D}_3 \rightarrow \mathcal{D}_1.$$

where $\mathcal{D}_3 = C^{1,2}([0, T] \times \mathbb{T}^3, \mathbb{R}^3)$ and $\mathcal{D}_1 = C^{1/2}([0, T], \mathcal{P})$

Given $v(x, t)$ from $\mathcal{D}_3$, the initial probability distribution μ is evolved according to its stochastic differential equation:

$$X_t = X_0 - \int_0^t v(s, X_s) ds + W_t,$$

where the distribution of X_0 is μ

Then define $\Phi_3^\mu[v]$ as the distribution of X_t

Lemma 2

The operator Φ_3^μ is a well-defined mapping from D_3 to D_1 satisfying the following estimates:

$$\exists \lambda \in \Lambda \text{ such that } \|\Phi_3^\mu[v]\|_{1T,1/2} \leq \lambda(|v|_0), \quad \forall v \in D_3,$$

and

$$d_{1T}(\Phi_3^\mu[v_1], \Phi_3^\mu[v_2]) \leq \lambda(|v_1|_{0,1} \vee |v_2|_{0,1})|v_1 - v_2|_0.$$

Furthermore, $\rho(t, x) = \Phi_3^\mu[v](t, x)$ is the unique classical solution of the Fokker–Planck equation:

$$\partial_t \rho - \nabla \cdot (\rho v) - \frac{1}{2} \Delta \rho = 0, \quad \rho(0, x) = \mu(x)$$

Lemma 1 gives a uniform bound on $|v|_{0,1}$. Therefore, the estimate in Lemma 2 yields continuity of Φ_3^μ on the relevant range of Φ_2 .

Fixed Point as a Solution

A valid solution must be a classical solution to the following modified Navier–Stokes system:

$$\begin{cases} (\partial_t + v \cdot \nabla)v &= -\nabla p[\rho] + \frac{1}{2}\Delta v, & (t, x) \in [0, T] \times \mathbb{T}^3, \\ \partial_t \rho + \nabla \cdot (\rho v) + \frac{1}{2}\Delta \rho &= 0, & (t, x) \in [0, T] \times \mathbb{T}^3, \\ v(0, x) &= \nabla h[\rho](x), & x \in \mathbb{T}^3, \\ \rho(T, x) &= \mu(x), & x \in \mathbb{T}^3. \end{cases}$$

Suppose $\rho(x, t)$ is a fixed point of Φ^μ

By the verification theorem, there exists a classical solution $v(x, t)$ to

$$(\partial_t + v \cdot \nabla)v = -\nabla p[\rho] + \frac{1}{2}\Delta v, \quad v(0, x) = \nabla h[\rho](x)$$

By Lemma 2, $\rho(x, t)$ is the unique classical solution to

$$\partial_t \rho + \nabla \cdot (\rho v) + \frac{1}{2}\Delta \rho = 0, \quad \rho(T, x) = \mu(x)$$

The Schauder Strategy

Theorem (Schauder Fixed Point)

Let K be a nonempty, compact, convex subset of a Banach space. If

$$F : K \rightarrow K$$

is continuous, then F has a fixed point.

To apply the Schauder Fixed Point Theorem, we need:

- 1 The map $\Phi^\mu = \Phi_3^\mu \circ \Phi_2 \circ \Phi_1$ to be continuous.
- 2 There is a nonempty, compact, and convex subset $\Gamma \subseteq \mathcal{D}_1$ such that $\Phi^\mu(\Gamma) \subseteq \Gamma$

Finishing the Proof

- ① Since Φ_3^μ , Φ_2 , and Φ_1 are all continuous, the composition $\Phi^\mu = \Phi_3^\mu \circ \Phi_2 \circ \Phi_1$ is continuous
- ② By the conditions of the theorem, $\exists \kappa > 0$ such that $|p[\rho]|_{0,2} + |h[\rho]|_4 \leq \kappa$ for all $\rho \in \mathcal{D}_1$

Lemma 2 implies that $|\Phi^\mu[\rho]|_{1T,1/2} \leq \lambda_1(|v|_0)$ for some $\lambda_1 \in \Lambda$

Lemma 1 implies that $|v|_0 \leq |v|_{0,1} \leq \lambda_2(\max\{|p[\rho]|_{0,2}, |h[\rho]|_4\}) \leq \lambda_2(\kappa)$ for some $\lambda_2 \in \Lambda$

Thus, $|\Phi^\mu[\rho]|_{1T,1/2} \leq \lambda_1(\lambda_2(\kappa))$

Define $\Gamma = \{\rho \in \mathcal{D}_1 : |\rho|_{1T,1/2} \leq \lambda_1(\lambda_2(\kappa))\}$

By the results above, we have

$$\Phi^\mu(\mathcal{D}_1) \subseteq \Gamma$$

Finishing the Proof

Moreover:

- Constant measure flow $\bar{\delta}_0 \in \Gamma$, so Γ is nonempty
- Γ is convex as convex combinations of Γ elements preserve $|\rho|_{1T,1/2} \leq \lambda_1(\lambda_2(\kappa))$
- Since $\mathcal{P}(\mathbb{T}^3)$ is compact and the elements of Γ are uniformly $\frac{1}{2}$ -Hölder continuous, the Arzelà–Ascoli theorem implies that Γ is relatively compact with respect to the d_{1T} metric.
- Since Γ is closed under convergence in the d_{1T} metric, it follows that Γ is compact.

By Schauder's fixed point theorem, there exists $\rho \in \Gamma$ such that:

$$\rho = \Phi^\mu[\rho].$$

Main takeaway:

The paper shows that a class of Navier–Stokes-type PDE systems can be transformed into a Mean Field Games system

We can prove the existence of a solution to the Mean Field Games system by expressing the system as an iteration and using Schauder's Fixed Point theorem

This Mean Field Games system can be transformed back into a classical solution to the considered Navier–Stokes-type PDE System

Thank You

Any Questions?